

The Complexity of Osteoarthritis: Interdisciplinary Insights and the Path to Modular Treatment

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Table of Contents

Section I: The Osteoarthritis Problem

I. Introduction

II. Scope of Osteoarthritis

- A. U.S. & International Effect
- B. Demographic Trends
- C. Financial Burden

III. Medical Diagnosis of Osteoarthritis

- A. Clinical Development
- B. Clinical Pain Response
- C. Strengths & Limitations of Brace Treatment

IV. Socio-Economic Analysis

- A. Socio-Economic Disparities in Treatment Access
- B. Implications of Socio-Economic Disparities in Treatment Access

Section II: Treatment Solution for Equitable Care

V. Solution

- A. Cross-Disciplinary Limitations in Brace Treatments
- B. Modular Brace Solution

VI. Conclusion

VII. Bibliography

VIII. Appendix

Section I: The Osteoarthritis Problem

I. Introduction

Osteoarthritis is a widespread and growing condition that affects millions globally, but its burden is not experienced equally. Osteoarthritis results in progressive degeneration of the cartilage, resulting in inflammation, pain, and motion restriction. The condition affects different individuals to a varied degree; for some, the pain is manageable and causes minor changes in habit, while for others it results in unbearable pain and entire lifestyle changes. Factors such as age, gender, geography, and socioeconomic status influence both the development of the condition and access to effective treatment. While braces are a non-invasive and often effective method for managing Osteoarthritis, current designs fall short in affordability, accessibility, and adaptability. The following discussion aims to educate readers on the need for modular and accessible brace treatments by examining the medical, demographic, and socioeconomic factors that shape the Osteoarthritis experience. This paper explores the growing burden of Osteoarthritis, beginning with an overview of its prevalence in the U.S. and internationally, as well as demographic trends and the significant financial strain it places on individuals. It then examines how Osteoarthritis is diagnosed, the biological mechanisms behind pain, and the current strengths and shortcomings of brace treatments. Following this, the paper analyzes socio-economic disparities that influence access to care and discusses their broader implications for treatment equity. In response to a holistic view of the Osteoarthritis problem, a solution is proposed that addresses cross-disciplinary limitations in existing brace treatments through the development of a modular brace. By addressing the Osteoarthritis problem through an interdisciplinary perspective that takes into account the clinical and social aspects, brace treatments have the potential to become systematic solutions.

II. Scope of Osteoarthritis

A. U.S. & International Effect

Osteoarthritis is a relevant problem that is becoming increasingly more prevalent, both in the United States and Worldwide. In the last five years, 53.2 million U.S. adults suffer from

arthritis, with 32.5 million of those suffering from Osteoarthritis.^{1,2} As average lifespan increases, the number of adults who suffer from arthritis will continue to grow. By 2040, it is projected that an estimated 78 million adults will suffer from arthritis.³ The prevalence and scope of Osteoarthritis extends beyond the United States; in 2019, a study found that 528 million people were diagnosed with Osteoarthritis worldwide. The upward trend for U.S. Osteoarthritis predictions is also reflected worldwide— the affected population found in 2019 had increased 113% since 1990.⁴

Globally, Osteoarthritis parallels similar trends. The GBG 2021 Osteoarthritis Collaborators researched the global, regional, and national burden of Osteoarthritis from 1990-2020, as well as computed projections to 2050.⁵ Osteoarthritis was studied in 204 nations and territories through the lens of geographies, age, sex, and time to estimate the impact of Osteoarthritis in the hand, hip, knee, and other locations. The study found that about 7.6% of the population was diagnosed with Osteoarthritis in 2020; the total number of diagnoses had increased by 132.2% since 1990.⁶ Further, the study proposed that diagnosis numbers would continue to increase. The expected overall rise in Osteoarthritis correlates an expected rise in average “years lived with disability”. The study attributes the overall rise in numbers to growing populations and age expectancy, rather than any particular factor making people more susceptible to Osteoarthritis.⁷ Growing numbers of Osteoarthritis diagnoses can be decomposed by demographic trends.

B. Demographic Trends

The overarching demographic factors that determine the prevalence of Osteoarthritis by demographic are age and gender. The prevalence of Osteoarthritis within a demographic

¹ Fallon, Elizabeth A., Michael A. Boring, Alyssa L. Foster, et al., “Prevalence of Diagnosed Arthritis — United States, 2019–2021,” *MMWR. Morbidity and Mortality Weekly Report* 72 (2023): 1101–1107.

² Centers for Disease Control and Prevention, *Osteoarthritis*, <https://www.cdc.gov/arthritis/Osteoarthritis/index.html> (accessed January 29, 2025).

³ Jennifer M. Hootman, Charles G. Helmick, Kayla E. Barbour, Kristina A. Theis, and Michael A. Boring, “Updated Projected Prevalence of Self-Reported Doctor-Diagnosed Arthritis and Arthritis-Attributable Activity Limitation among US Adults, 2015–2040,” *Arthritis & Rheumatology* 68, no. 7 (2016): 1582–1587, <https://doi.org/10.1002/art.39692>.

⁴GBD 2019, *Global Burden of 369 Diseases and Injuries in 204 Countries and Territories, 1990–2019: A Systematic Analysis for the Global Burden of Disease Study 2019*, <https://vizhub.healthdata.org/gbd-results/> (accessed 5/3/2025).

⁵ Chiropractic Resource Organization, “Burden of Osteoarthritis: Global, Regional, and National Trends, 1990–2020 and Projections to 2050,” *Chiro.org*, https://chiro.org/Global_Burden/Burden_of_Osteoarthritis.shtml (accessed April 16, 2025).

⁶ Chiropractic Resource Organization, “Burden of Osteoarthritis.”

⁷ Chiropractic Resource Organization, “Burden of Osteoarthritis.”

increases with age. The CDC produced research in 2024 that found 3.6% of adults aged 18-34 had Osteoarthritis whereas 53.9% of adults 75 and older had Osteoarthritis.⁸ This trend is reflected globally, and contributes to the predicted rise in Osteoarthritis numbers worldwide. Along with age, gender is a significant factor contributing to susceptibility. Figure 3 graphs the prevalence of Osteoarthritis depending on age and time, and visually captures the relationship between those two factors. Osteoarthritis increases in a similar pattern for both genders as they age, though the numbers are higher following this pattern for women than for men. Further, Figure 4 expands the previous figure, focusing on the most common locations for Osteoarthritis. All four of these plots show the male and female numbers following similar patterns, with women more likely to be diagnosed. The four plots convey an additional trend connecting location of Osteoarthritis and time. Locations of Osteoarthritis in the hand, hip, and other areas seem to follow similar, relatively linear patterns where the number of people diagnosed increases directly with age. On the other hand, Osteoarthritis in the knee seems to model a relationship roughly parabolic, where the highest numbers of diagnoses seem to increase until around 80 years of age, and then decrease for ages older than 80. The scope of Osteoarthritis is most prevalent in certain demographics, but has a broad relevance and burden across population groups.

C. Financial Burden

The immense financial burden of Osteoarthritis directly follows its widespread prevalence. There is both a direct and indirect financial burden of Osteoarthritis, resulting from the medical condition itself as well as its widespread effect on people's lives.

In 2020, Osteoarthritis was a top-ten leading cause of years lived with disability for the adult demographic over the age of 70, and continues to likewise be a cause of disability among working-aged adults.⁹ Indirect costs of the condition often arise from the 1) lost pay due to forced job changes and absenteeism, 2) the cost of treating health issues divergent from Osteoarthritis. Within the United States, collectively there was an estimated \$164 billion lost in wages due to Osteoarthritis in 2013; the increase in Osteoarthritis since as well as project trends support the increasing relevance of the economic burden.¹⁰ Physical limitations,

⁸ National Center for Health Statistics, *Arthritis in Adults: United States, 2021–2022*, CDC Data Brief No. 497, April 2024, <https://www.cdc.gov/nchs/products/databriefs/db497.htm> (accessed April 29, 2025).

⁹ Chiropractic Resource Organization, "Burden of Osteoarthritis."

¹⁰ Osteoarthritis Action Alliance, "OA Prevalence and Burden," *OA Module*, <https://oaaction.unc.edu/oa-module/oa-prevalence-and-burden/> (accessed April 29, 2025).

comorbid conditions, and medication side effects increase overall mortality rate by 55% overall increasing the financial burden of medical and other resulting costs.¹¹

Directly, the annual cost of arthritis averages to around \$11,000 per person.¹² The cost of treating the condition directly varies dramatically, as do treatment results. For example, joint injections are very effective for some and offer no relief for others. Depending on the severity of the condition, treatments might be costly but structurally alleviate the issue: such as joint replacements and experimental therapies. Further, cheaper methods of treating the condition often do not stand alone and are used in conjunction with one another, combining pain medication with brace use for example. Table 1 outlines the most common treatment options for Osteoarthritis with expected price ranges per treatment. Another factor that affects the financial burden is the insurance coverage of treatment options. Generally, insurance covers physical therapy, prescription medications, braces, and joint replacement surgery. However, the amount covered varies depending on the plan, and often requires the deductible be met before coverage. As the burden of Osteoarthritis often extends beyond the immediate discomfort an individual experiences, the burden is often multifaceted and vast.

III. Osteoarthritis: Medical Diagnosis of Osteoarthritis

A. Clinical Development

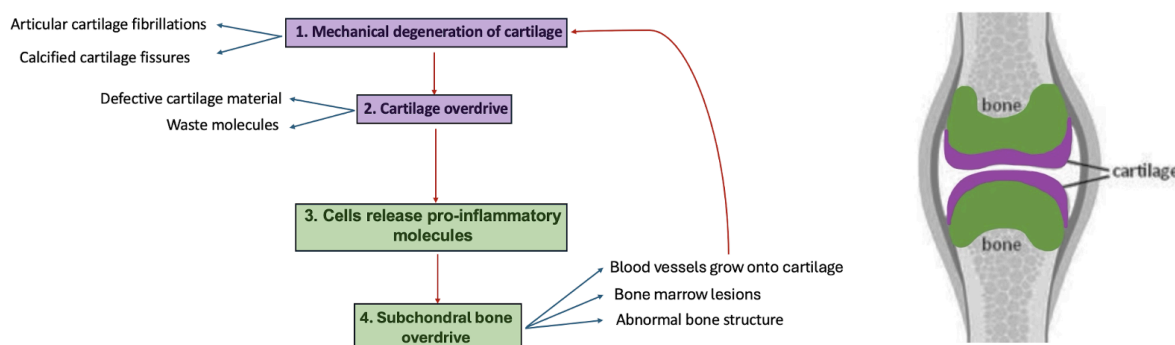


Figure 1: Development Feedback Loop & Associated Joint Location

¹¹ Osteoarthritis Action Alliance, "OA Prevalence and Burden."

¹² Osteoarthritis Action Alliance, "OA Prevalence and Burden."

Osteoarthritis development is iterative and heterogenous. Osteoarthritis is medically defined as the mechanical breakdown of cartilage due to long-term use or injury, resulting in bone-on-bone contact that initially develops from mechanical wear and tear or injury.¹³ Osteoarthritis is a specific subcategory of arthritis along with Rheumatoid arthritis, which differ in both cause and development. Rheumatoid arthritis is an autoimmune condition in which the immune system mistakenly attacks the synovial membrane—the lining of the joints—causing inflammation, pain, and joint damage.¹⁴ The development of the condition also differs; Rheumatoid arthritis affects joints symmetrically, whereas Osteoarthritis only affects the “overused” joints that have developed the condition which can cause an asymmetrical development. Diagnosing the specific condition responsible for a person’s joint pain can be difficult, as a certain condition can manifest differently person to person, but ultimately there are developmental and physiological differences within Arthritis, as shown in Table 2.

Though the development of Osteoarthritis can be generally described as mechanical degeneration, the medical pathogenesis is more nuanced. The process can be characterized as a biological feedback loop that involves the entire joint, where the pain and inflammation are both results and perpetrators of the condition. Though often thought of as degenerating passively, the degeneration is dynamic. Cycles of mechanical, inflammatory, and metabolic factors result in eventual structural failure of the synovial joint.¹⁵ The process mainly originates in the bone and cartilage of the joint, and can be generally categorized in four steps in reference to Figure 1.

1. Mechanical degeneration of cartilage¹⁶

The cartilage between the joints changes in composition, making it more susceptible to physical effects. The result is that movements originally supported by a joint, are now abrasive. The mechanical degeneration of the cartilage begins as fibrillations—the shredding or fraying of the articular cartilage tissue that covers the end of bones where they meet at a joint. Cartilage damage can expand to deep cartilage fissures in the calcified cartilage.

¹³ Mayo Clinic Staff, “Osteoarthritis – Symptoms and Causes,” *Mayo Clinic*, October 4, 2023, <https://www.mayoclinic.org/diseases-conditions/Osteoarthritis/symptoms-causes/syc-20351925> (accessed April 29, 2025).

¹⁴ Z. H. Alsous, “What’s the Difference Between Rheumatoid Arthritis and Osteoarthritis?,” *Medical News Today*, April 9, 2024, <https://www.medicalnewstoday.com/articles/323617> (accessed April 29, 2025).

¹⁵ David J. Hunter and Sita Bierma-Zeinstra, “Osteoarthritis,” *The Lancet* 393, no. 10182 (2019): 1745–1759, <https://www.sciencedirect.com/science/article/abs/pii/S0140673619304179> (accessed April 29, 2025).

¹⁶ Hunter and Bierma-Zeinstra, “Osteoarthritis,” *The Lancet*, 1745–1759.

2. Cartilage overdrive¹⁷

The body responds to the damage by enlarging the cartilage cells, which overproduce cartilage material. Normal chondrocyte cells enlarge to hypertrophic chondrocyte cells and overdrive production to “fix” the cartilage damage. This process produces waste and signals of inflammation as byproducts, which 1) changes the normal state of cartilage cells and cartilage development process, 2) stimulates the synovium joint lining. In response to the stimulation, the cells in the synovium, synoviocytes, multiply and grow in overdrive, releasing inflammatory substances as a byproduct. The stimulated synovium results in the thickening of the joint lining and increased blood flow; both of which directly increase or lead to the production of inflammation.

3. Cells release proinflammatory molecules¹⁸

This breakdown process of the cartilage produces proinflammatory signals, which send the bone into overdrive production.

4. Subchondral bone overdrive¹⁹

Overproduction in the bone results in long-term damage. The bone may thicken, form bone spurs, bruise areas inside the bone, and allow new blood vessels to grow onto cartilage. The growth response results in inflammation, joint stress, and poor movement patterns. The degeneration of the cartilage also decreases the absorption and cushioning effect of the joint and displaces the pressure on the bone. The layer of the bone right under the cartilage— the subchondral bone— is subjected to greater stress than normal, and reacts by breaking down old tissue and making new bone quicker than usual. In addition, the proinflammatory signals produced due to cartilage breakdown additionally prompt the subchondral bone development into overdrive.²⁰ The heterogenous and progressive development of Osteoarthritis produces a similar pain response system.

¹⁷Hunter and Bierma-Zeinstra, “Osteoarthritis,” *The Lancet*, 1745–1759.

¹⁸Hunter and Bierma-Zeinstra, “Osteoarthritis,” *The Lancet*, 1745–1759.

¹⁹Hunter and Bierma-Zeinstra, “Osteoarthritis,” *The Lancet*, 1745–1759.

²⁰ Hunter and Bierma-Zeinstra, “Osteoarthritis,” 1745–1759.

B. Clinical Pain Response

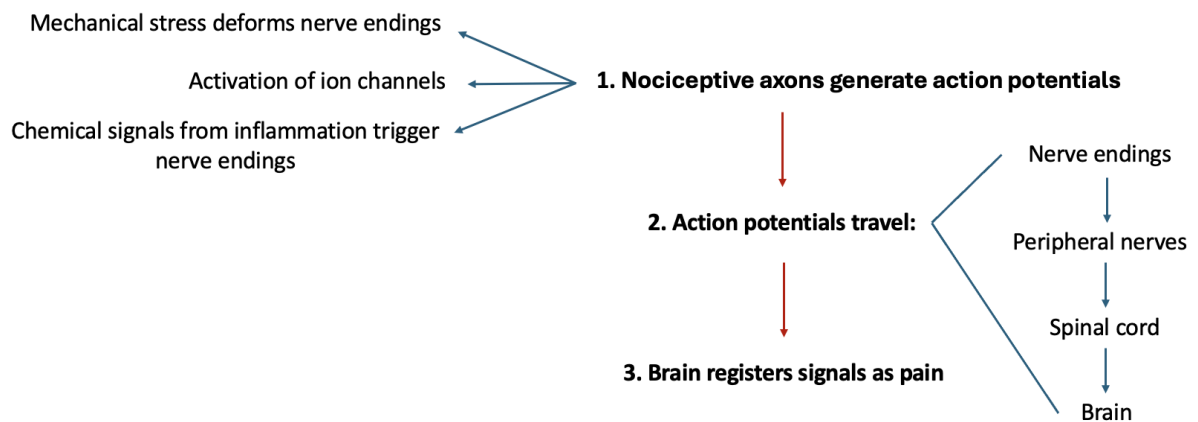


Figure 2: Pain Response Process

In Osteoarthritis, the degeneration of cartilage results in bone-on-bone contact. The pain does not arise directly from this contact, but as a result of the nerve endings inside the bone reacting to damage. The process can be described generally in three parts in reference to Figure 2:

1. Nociceptive axons generate action potentials²¹

The damage from the cartilage degeneration causes chemical and mechanical pain triggers. Chemically, the same pro-inflammatory molecules released in the degeneration process sensitize the nociceptors, making them more easily triggered. Mechanically, the physical damage to the subchondral bone activates the nociceptors, which are triggered by stretching and compression of the tissue they are within. Once triggered, the nociceptors produce an electric signal called an action potential. The ion channels are also activated by these chemical and mechanical triggers, and open—allowing the action potential to travel to the brain and register pain. The dynamic and effect of the chemical and mechanical factors are similar to a broken house. As the house breaks down, chemicals from the paint and other materials are released, and the physical structure of the house simultaneously degenerates. Throughout this process, the chemical and motion sensors become increasingly sensitive; when

²¹ R. C. de Souza, C. G. B. Lopes, M. A. T. de Carvalho, and T. A. A. Rodrigues, "Understanding the Pain Mechanism in Osteoarthritis: A Scoping Review," *Osteoarthritis and Cartilage Open* 5, no. 3 (2023): 100387, <https://www.sciencedirect.com/science/article/pii/S2772594423002133> (accessed April 29, 2025).

they are finally triggered, alarms blare and the doors open, theoretically allowing the owner to escape. In the bone, the chemical and physical sensors increase in sensitivity throughout the generation, and when the nociceptors are triggered, the ion channels open for the action potential to flow through.

2. Action potential travels to the brain²²

The action potential travels from the nerve endings in the bone, through the peripheral nerves and spinal cord, and to the brain through the ion channels.

3. The brain interprets the signal as pain²³

The action potential reaches the brain, where areas like the thalamus and somatosensory cortex decode the signal. The brain then interprets it as pain, giving meaning to the nerve activity based on its location, intensity, and context.²⁴

The heterogenous development and inconsistent pain response of Osteoarthritis require a modular and autonomous treatment solution.

C. Strengths & Clinical Limitations of Brace Treatment

Braces offer treatment in a multifaceted way that is non-invasive and autonomous. Doctors often study the CT or MRI scans of Osteoarthritis patients to research treatments for the pain. These scans show changes in bone structure, bone marrow, and inflammation but the level of damage is inconsistent with the felt-pain of a patient's experience. Since the level of damage is not necessarily indicative of the pain response, the current CT and MRI scans do not fully illuminate the condition in its entirety. Due to the heterogeneous Osteoarthritis development and its pain response, a brace solution that is modular in the level of support it provides would be an inclusive option to patients across the board. It would provide an established method to manage the pain, and works independently of any medical understanding of the pain level an individual is experiencing. Braces provide patients autonomy over their own pain treatment, as their felt experience of the condition supersedes accessible medical tests to determine pain levels.

²² de Souza et al., "Understanding the Pain Mechanism in Osteoarthritis," 100387.

²³ de Souza et al., "Understanding the Pain Mechanism in Osteoarthritis," 100387.

²⁴ D. C. Turk and R. Melzack, "The Anatomy and Physiology of Pain," in *Handbook of Pain Assessment*, 3rd ed. (New York: Guilford Press, 2011), 3–23, <https://www.ncbi.nlm.nih.gov/books/NBK219252/> (accessed April 29, 2025).

Braces manage the mechanical and chemical components of Osteoarthritis development and pain response. Mechanically, underloading braces help to shift the weight away from the damaged joint which can slow further cartilage degeneration and reduce inflammation. Further, the joint alignment provides support that prevents abnormal joint movements that are often a side effect of managing the pain and inflammation. This mechanically can contribute to slower cartilage degeneration and reduced inflammation. Braces manage the chemical response of the pain signals in the joint by limiting painful motion and reducing pressure on inflamed areas. Braces offer an autonomous and non-invasive treatment solution that aids in the development and pain response of the condition, though brace treatment can be pricey and hard to obtain due to systemic factors affecting the accessibility of healthcare.

IV. Socio-Economic analysis

A. Socio-Economic Disparities in Treatment Access

To fully understand the scope of Osteoarthritis, the condition and its influences need to be considered in socio-economic context. Both the development and burden of the condition is complex. Research published through the Centers for Disease Control and Prevention (CDC) used data from the National Health Interview Survey from 2002, 2003, and 2006 to investigate the correlation of race with susceptibility to develop arthritis and the impact of the condition.²⁵ The study found that within the United States, Hispanics and Blacks were less likely to develop arthritis, but were more likely to be impacted by the condition. Hispanics and Blacks who developed arthritis were 1.3 times as likely as Whites to experience activity limitation, 1.9 times as likely to experience severe joint pain, 1.7 times as likely to experience work restriction. Multiracial people were 1.9 times as likely as Whites to report activity limitation and severe joint pain, 1.9 times as likely to experience work limitation, and 2.2 times as likely to experience work restriction.²⁶ The dissonance between groups that are most susceptible to development and groups that are most impacted by the condition reflects the involved nature of the condition.

Socio-economic factors impact patients' access to surgical and non-surgical treatments. The most direct factor is cost, but geographical location and access to physician care are indirect but significant factors. Individuals with lower socioeconomic status are more likely to face transportation barriers, which lead to delayed or missed medical care, and worsening

²⁵ Jennifer Bolen, Linda Schieb, Jennifer M. Hootman, et al., "Differences in the Prevalence and Severity of Arthritis among Racial/Ethnic Groups in the United States, National Health Interview Survey, 2002, 2003, and 2006," *Preventing Chronic Disease* 7, no. 3 (2010): A64.

²⁶ Bolen et al., "Differences in the Prevalence and Severity of Arthritis," A64.

health outcomes. These barriers disproportionately affect underinsured and low-income patients, contributing to systemic healthcare inequities and higher long-term healthcare costs.²⁷ Further, a study found that patients with Medicaid insurance had limited access to care in 32% of orthopedic practices, with only 36% being offered an appointment within two weeks compared to 89% for those with private insurance. Patients with Medicaid insurance were more likely to travel longer distances for orthopedic appointments and had higher missed appointment rates, indicating significant barriers to accessing specialized care.²⁸ Even with access to physician care, physician bias provides an additional barrier to Osteoarthritis care. Research shows that Black patients are referred for Osteoarthritis surgery less frequently than White patients and are more often directed to smaller provider networks.²⁹ Again, insurance type and geographic location are overlapping factors with socio-economic status that emphasize the systemic nature of accessibility within Osteoarthritis treatment. Socio-economic factors of affordability and physician bias greatly impact the accessibility of Osteoarthritis treatment.

B. Implications of Socio-Economic Disparities in Treatment Access

A similar socio-economic trend is reflected in patients' access to Osteoarthritis treatments, suggesting a correlation between access to treatment and the lived burden of the condition. Joint replacement surgeries are as close as a cure one can get for Osteoarthritis, and access to such treatment significantly decreases the lifestyle burden of the condition. Within this treatment, socio-economic factors directly correlate with the rate at which joint replacement treatments are utilized. Individuals with higher income, education, and private insurance— a demographic disproportionately represented by White patients— are more likely to undergo joint replacement surgeries.³⁰ Access barriers are often more nuanced than simply economic, but the result is the same; people from lower socioeconomic backgrounds are less likely to be referred

²⁷ A. Syed, H. M. Gerber, R. M. Keane, and A. T. Culyba, "Transportation Barriers to Healthcare in the United States: Findings from the National Health Interview Survey, 1997–2017," *BMC Public Health* 22, no. 1 (November 2022): 2052, <https://bmcpublichealth.biomedcentral.com/articles/10.1186/s12889-022-14149-x> (accessed April 29, 2025).

²⁸ B. Labrum, S. M. Iorio, M. K. Dines, J. T. Nguyen, and C. G. Haskell, "Insurance Status Affects Access to Orthopedic Care: A Comparison of Medicaid and Private Insurance," *Journal of the American Academy of Orthopaedic Surgeons* 28, no. 15 (August 2020): e646–e651, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7406150/> (accessed April 29, 2025).

²⁹ C. Bailey, "Racism & Inequality in Healthcare for African Americans," *The Century Foundation*, February 11, 2020, <https://tcf.org/content/report/racism-inequality-health-care-african-americans/> (accessed April 29, 2025).

³⁰ C. J. Holtzman, A. S. Newcomb, K. R. Wilson, and C. M. Kraus, "Racial Disparities in Osteoarthritis: A Systematic Review of Risk Factors," *BMC Rheumatology* 4, no. 1 (April 2020): 17, <https://bmcrheumatol.biomedcentral.com/articles/10.1186/s41927-020-0117-0> (accessed April 29, 2025).

for or receive these procedures, even when the clinical need is equal to or greater than others with higher socioeconomic status. Even within the category of joint replacement treatments, there are disparities in the quality of treatment received due to price, physician bias, and health literacy. Two common knee surgery treatments are unicompartmental knee arthroplasty (UKA) and total knee arthroplasty (TKA). UKA is associated with lower morbidity than TKA, though UKA treatments are disproportionately utilized by White patients, higher income patients, and privately insured patients compared to Black, lower income, and Medicare insured patients.³¹ It is worth noting that these factors are race biased themselves; Black patients are more likely to be lower income and Medicare insured, whereas White patients are more likely to be higher income and privately insured.³² Access to Osteoarthritis treatment is limited by economic, geographic, and physical barriers, disproportionately affecting lower socioeconomic groups and leading to a higher burden of the disease in these populations.

Section II: Treatment Solution for Equitable Care

V. Solution

A. Cross-Disciplinary Limitations in Brace Treatments

The problem of Osteoarthritis spans multiple disciplines. Its impact is significant both nationally and globally, with projections indicating that its prevalence will continue to rise. Medically, Osteoarthritis develops in varied ways and produces a pain response that is challenging to interpret and manage. Because the lived experience of Osteoarthritis often depends on the availability and quality of treatment, access to medical care plays a crucial role in shaping outcomes. Individuals with lower socioeconomic status face greater financial obstacles, limited access to care, and provider bias, all of which contribute to a higher overall burden of the condition.

A modular approach is needed to address the complex and multifaceted issues of Osteoarthritis. Clinically, braces mitigate the heterogeneous development and inability to precisely monitor and model pain response. They help manage the condition's progression

³¹ S. A. Dhanjani, J. Schmerler, N. Hussain, D. Badin, U. Srikumaran, V. Hegde, and J. K. Oni, "Racial, Socioeconomic, and Payer Status Disparities in Utilization of Unicompartmental Knee Arthroplasty in the USA," *Knee Surgery & Related Research* 37, no. 1 (December 2025): 2, <https://doi.org/10.1186/s43019-024-00227-4> (accessed April 29, 2025).

³² Bailey, "Racism & Inequality in Healthcare."

through kinesiological support and manage its complex pain response by mitigating progression and inflammation. Further, they are non-invasive, autonomous, and generally more affordable than surgical or pharmaceutical alternatives.

However, current brace designs fall short in addressing the full scope of the problem. While braces may be less expensive than other treatments, they are still not financially accessible for everyone, particularly those with limited income. Since Osteoarthritis often affects multiple joints in the body, purchasing a separate brace for each affected area can create additional financial strain. Many specialized braces also require fitting by a physician, which introduces further barriers for individuals who may face challenges related to geography, cost, or provider discrimination. There is a clear need for modular brace solutions, as modularity, affordability, and accessibility go hand in hand.

B. Modular Brace Solution

A brace designed to adapt to multiple body locations and offer various levels of support would significantly reduce costs by combining the functions of several braces into a single product. One potential solution is a wrap-style design, where users apply a base brace and layer additional wrapping according to specific patterns tailored to different joints and levels of support. A potential sketch of such a brace is shown in Figure 5, where the same brace adapts to both the knee and wrist joint. The base sleeve is shown in red, and the wrap is shown in green—though wrap patterns would appear differently depending on specified level of support. The design for the wrap pattern would be inspired by kinesiological research, which would allow cheaper fabrics to be used in a way that provides comparable support to more advanced materials at a fraction of the cost. The basic kinesiological design corresponding to Figure 5 is shown in Figure 6. Such a design would allow individuals to customize their treatment independently and more affordably, while also reducing reliance on medical professionals for fitting. Further, this type of design would also be more inclusive, accommodating a wide range of pain levels and condition development, both within one person over time and across different individuals. By addressing both financial limitations and the medical heterogeneity of Osteoarthritis, this solution could effectively serve a broader population and meet the varied needs associated with the condition.

In addition to affordability, accessibility is essential. To overcome geographic challenges and reduce the need for physician referrals, braces should be made available through widely used retail chains such as Walmart and Walgreens. Research should focus on underserved communities to determine which stores are already part of their daily routines to integrate brace

distribution with their habits. For example, if individuals frequently shop for groceries at a local Walmart, that store could serve as an ideal location for brace distribution. This would help eliminate the need for extra appointments, transportation, or searching for medical providers, and would empower individuals to manage their treatment more independently. The ability to adjust the modular brace into different configurations for personalized support gives users even more control to treat their condition in a non-invasive manner.

VI. Conclusion

The burden of Osteoarthritis is rooted in systemic issues. The condition itself is not always inherently disabling; rather the disparity in its impact often stems from unmet needs in the design and delivery of treatment. For those with consistent access to quality care and treatment, Osteoarthritis can be effectively managed, significantly reducing its burden. However, for many others, specifically individuals with lower socioeconomic status, barriers to care dramatically increase the burden of the condition.

Much of the suffering associated with Osteoarthritis is not solely a result of the disease, but evolves from barriers that prevent effective treatment. These barriers include cost, insurance limitations, geographic inaccessibility, and physician bias. As a result, developing effective treatment options must go beyond strictly medical functionality and must consider accessibility, inclusivity, and affordability.

Braces offer a promising non-invasive solution that can support joint stability and manage pain. However, current brace designs often fall short in meeting the needs of underserved populations. Most are not modular, requiring separate products for each affected joint, which increases cost. Many also require physician fitting, which introduces additional obstacles for those who cannot afford appointments or who may face physician discrimination.

To address the systemic nature of the problem, a systemic solution is required. The treatment must be designed through a comprehensive lens that considers not only the physical design of the brace, but also its affordability, accessibility, and distribution. A modular, self-configurable brace that can be sold through familiar and accessible retail locations would help overcome economic, geographical, and social barriers. Ultimately, designing for real-world access in addition to medical functionality is necessary in designing an effective Osteoarthritis treatment.

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VIII. Appendix

Table 1: Common Osteoarthritis Treatments & Prices^{33 34}

Treatment	Price range
Braces	\$50 - \$1000
Physical therapy	\$75 - \$150 per session
Over-the-Counter Medication	\$10 - \$30 per month
Prescription Medication	\$30 - \$100 per month
Injections	- Corticosteroid: \$100 - \$300 - Hyaluronic Acid: \$300 - \$1000
Experimental	- Stem Cell Therapy: \$3,000 - \$ 5,000 - Platelet-Rich Plasma Therapy: \$1,000 - \$2,000
Joint replacement	\$20,000 - \$30,000

Table 2: Diverging Behaviors in Arthritis Types

	Osteoarthritis	Rheumatoid Arthritis
Development	Gradual; Degenerative	Sudden; Autoimmune
Geometry	Asymmetric	Symmetric
Stiffness	Worse after use	Can improve with use

³³ National Health Service (NHS), "Osteoarthritis – Treatment," *NHS.uk*, <https://www.nhs.uk/conditions/Osteoarthritis/treatment/> (accessed April 17, 2025).

³⁴ Mayo Clinic Staff, "Osteoarthritis – Diagnosis & Treatment," *Mayo Clinic*, April 8, 2025, <https://www.mayoclinic.org/diseases-conditions/Osteoarthritis/diagnosis-treatment/drc-20351930> (accessed April 29, 2025). <https://www.mayoclinic.org/diseases-conditions/Osteoarthritis/diagnosis-treatment/drc-20351930>. [Accessed: Apr. 17, 2025]

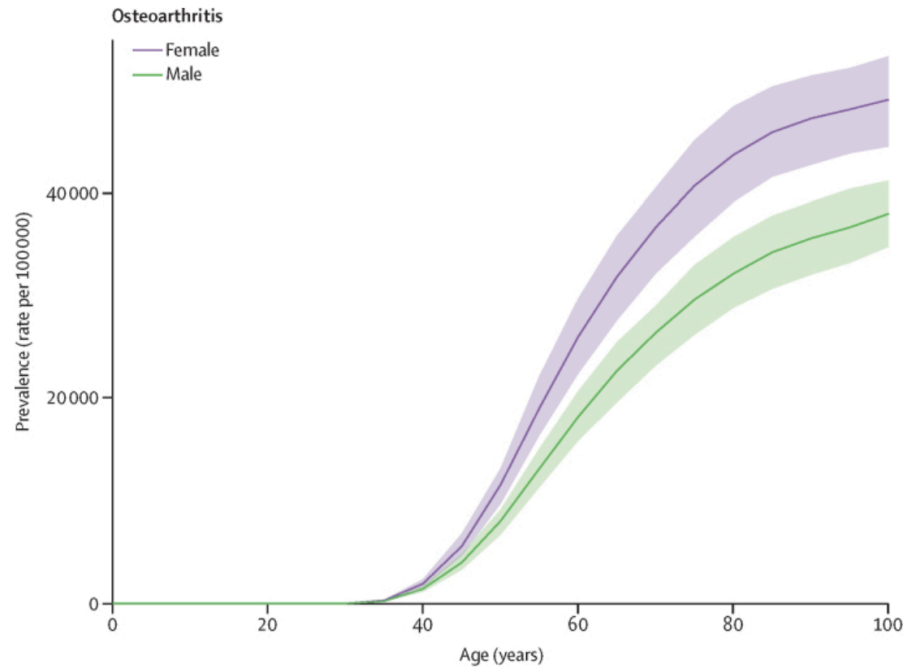


Figure 3: Osteoarthritis Prevalence (rate per 100,000) vs Age: by Gender³⁵

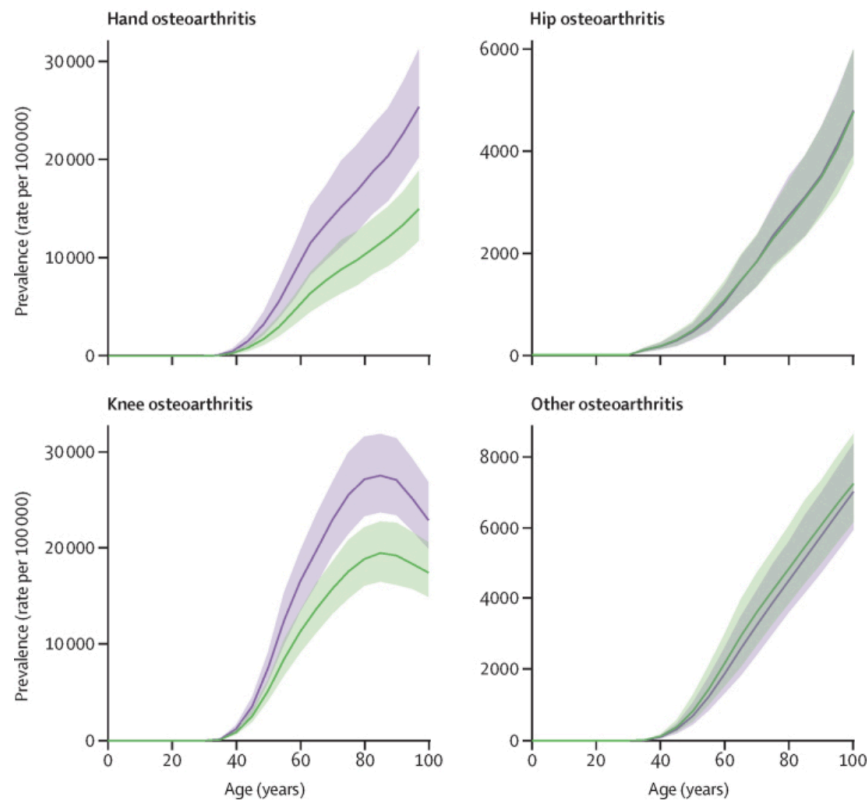


Figure 4: Osteoarthritis Prevalence (rate per 100,000) vs Age: by Gender & Type³⁶

³⁵ Chiropractic Resource Organization, "Burden of Osteoarthritis."

³⁶ Chiropractic Resource Organization, "Burden of Osteoarthritis."

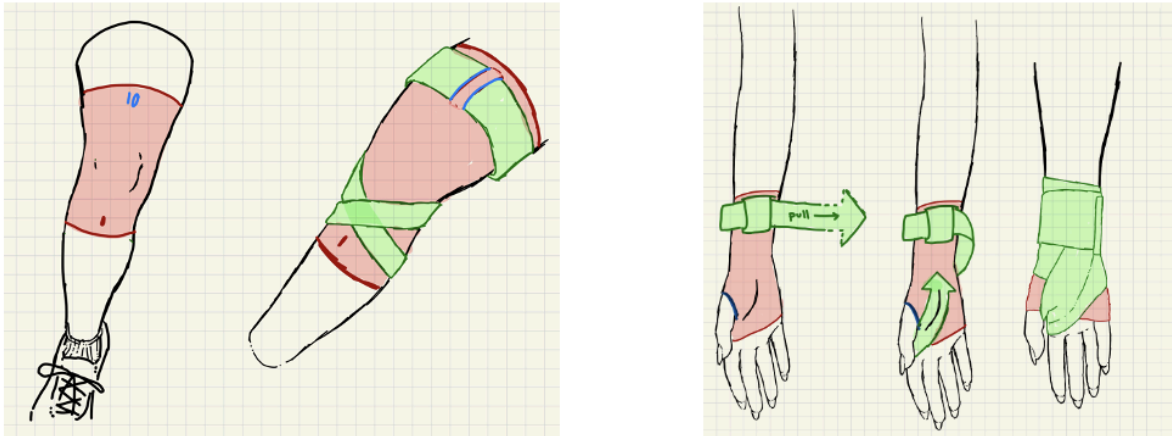


Figure 5: Base and Wrap Design for Knee/Wrist Brace

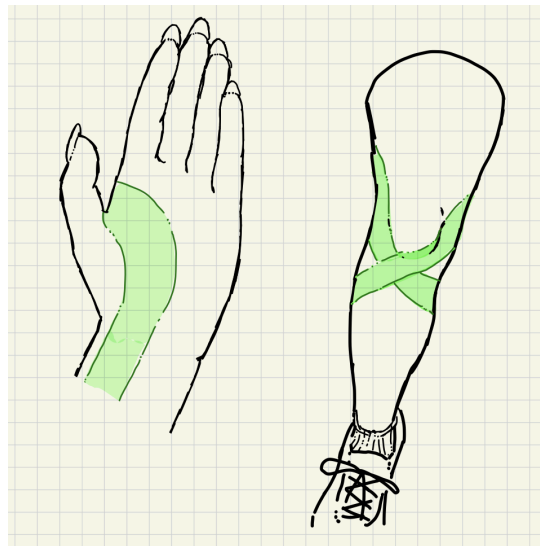


Figure 6: Foundation to Wrap Patterns